Engineering Research in Computer Systems (13-Week Syllabus)

Level: MS / early PhD **Focus:** Operating Systems, Networks, Computer Architecture, Security **Outcome:** A conference-ready systems paper + reproducible artifact

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Week 1 — What "Research" Means in Systems

Topics:

- Systems research vs. engineering
- Three pillars: performance, correctness, scalability
- Types of contributions: mechanism, insight, evaluation
- Storytelling in systems: problem → mechanism → design → evaluation

Readings:

- Saltzer & Kaashoek, *Principles of Computer System Design*, Ch. 1
- Hennessy & Patterson, "A New Golden Age for Computer Architecture" (CACM)
- David C. Clark et al., "Design Philosophy of the DARPA Internet Protocols"

Deliverable:

- Pick 3 SOSP/OSDI/NSDI papers
- Identify for each: problem, mechanism, system built, evaluation metric, why accepted/rejected

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Week 2 — How to Read and Analyze Systems Papers

Topics:

- Paper anatomy: problem → insight → system → evaluation
- Spotting novelty vs incremental work

Readings:

- Keshav, *How to Read a Paper*
- Arpaci-Dusseau & Arpaci-Dusseau, *Operating Systems: Three Easy Pieces*, Ch. 1
- Any recent OSDI/NSDI best paper

Deliverable:

- Dissect 1 paper: assumptions, system model, bottleneck, what could break

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Week 3 — Finding Research Problems

Topics:

- Sources of problems: hardware, workloads, failures, security attacks
- Recognizing publishable gaps

Readings:

- HotOS proceedings (last 2 editions)
- Hennessy & Patterson, *Computer Architecture: A Quantitative Approach*, Ch. 1
- Barroso et al., *The Datacenter as a Computer*, Ch. 1

Deliverable:

- Write 5 problem statements using:

> "Current systems fail when ___ because ___."

Week 4 — Turning Problems into Research Questions

Topics:

- Converting problems into measurable, falsifiable hypotheses
- Research question → hypothesis → metrics

Readings:

- Jim Gray, "Why Do Computers Stop and What Can Be Done About It?"
- Dean & Barroso, "The Tail at Scale" (CACM)
- One NSDI paper with strong evaluation

Deliverable:

- For 2 problems:

- Research question
- Hypothesis
- Metric

Week 5 — Experimental Methodology

Topics:

- Microbenchmarks vs macrobenchmarks
- Controlled experiments
- Threats to validity in systems experiments

Readings:

- Mytkowicz et al., “Producing Wrong Data Without Doing Anything Obviously Wrong”
- ACM SIGMETRICS evaluation guidelines
- Paxson, “Strategies for Sound Internet Measurement”

Deliverable:

- Design a full experiment: variables, controls, metrics, expected outcomes

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Week 6 — Systems Prototyping

Topics:

- Prototyping levels: kernel/user-space, emulator, trace-driven simulation
- Build only what touches the bottleneck
- Minimal prototypes vs full systems

Readings:

- Feamster et al., “The Case for Reproducible Networking Research”
- Arpaci-Dusseau, “The Development of the Cedar File System”
- EuroSys papers with new systems

Deliverable:

- Decide: real system, emulator, or trace-based model
- Justify choice

Week 7 — Measurement Infrastructure

Topics:

- Observing system behavior correctly
- Low-level vs high-level metrics
- Perf, eBPF, tracing, profiling

Readings:

- Brendan Gregg, *Systems Performance*, Ch. 1–3
- Linux perf documentation
- Gregg, "The USE Method"

Deliverable:

- Build measurement scripts producing latency distribution, CPU breakdown, queueing behavior

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Week 8 — Data Analysis for Systems

Topics:

- Tail latency, throughput, resource efficiency
- Repeatability, variance, confidence intervals
- Avoid cherry-picking

Readings:

- Dean & Barroso, "The Tail at Scale"
- Jain, *The Art of Computer Systems Performance Analysis*, Ch. 11
- SIGMETRICS papers

Deliverable:

- Graphs showing mechanism + performance: mean, P95, P99

- Explain why improvements occur

Week 9 — Positioning Against Prior Work

Topics:

- Literature mapping: table of prior systems
- Identifying gaps
- Related Work storytelling
- Anticipating reviewer critiques

Readings:

- Armstrong et al., "Towards a Common Benchmark for DB Systems"
- Feamster & Rexford, "Why (and How) Networks Should Run Themselves"
- Related Work sections from OSDI/SOSP papers

Deliverable:

- Literature map table
- Head-to-head comparison plan
- Draft Related Work section
- Gap statement

Week 10 — Writing the Paper

Topics:

- Paper structure: title, abstract, intro, evaluation, conclusion
- Storytelling through metrics
- Emphasize mechanism and impact

Readings:

- Simon Peyton Jones, "How to Write a Great Research Paper"
- OSDI paper template (USENIX)
- Top systems papers (OSDI/SOSP/NSDI)

Deliverable:

- Draft title, abstract, and introduction

Week 11 — Evaluation That Survives Reviewers

Topics:

- Ablation, sensitivity, stress tests
- Show mechanism behind performance improvements
- Edge cases (P50, P95, P99)
- Repeatability and statistical rigor

Readings:

- Pat Helland, “Life Beyond Distributed Transactions”
- Harchol-Balter, *Performance Modeling and Design of Computer Systems*, Ch. 2
- 2 NSDI papers with deep evaluation

Deliverable:

- Complete evaluation plan with:
  - Ablation and sensitivity studies
  - Head-to-head comparisons
  - Mechanism graphs
  - Repeatable scripts

Week 12 — Reproducibility and Artifacts

Topics:

- Docker, scripts, VMs for reproducible experiments
- Artifact evaluation committees
- Open science and sharing

Readings:

- NSDI Artifact Evaluation Guidelines
- “Repeatability in Computer Systems Research” (EuroSys)
- Docker / containerization documentation

Deliverable:

- Package system and experiments so another student can run them successfully

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Week 13 — Submission and Review

Topics:

- How top-tier systems conference review works
- Rebuttal strategies
- Surviving rejection and revision

Readings:

- “How to Get a Paper Accepted at OSDI” (Stoica)
- SIGCOMM / SOSP review guidelines
- Sample HotCRP reviews

Deliverable:

- Mock program committee: review classmates’ papers
- Prepare rebuttal for your own draft

Final Outcome

By the end of Week 13, you will have:

- A **working systems prototype**
- A **real evaluation with graphs, stats, and comparisons**
- A **conference-ready paper** for OSDI/SOSP/NSDI/EuroSys

This syllabus follows the same pipeline used in **MIT PDOS, Berkeley RISELab, CMU Systems, ETH Systems, and Microsoft Research**.